

MTH-ALG, Algebra Problem Sets Solutions

ATONU ROY CHOWDHURY, achowdhu@ictp.it

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§1 Problem Set 1

Exercise 1.1. Let

$$G = \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/20\mathbb{Z}. \quad (1.1)$$

- (a) How many elements of G have order 60?
- (b) How many cyclic subgroups of G have order 30?
- (c) How many injective homomorphisms

$$f : \mathbb{Z}/12\mathbb{Z} \longrightarrow G \quad (1.2)$$

are there?

Solution. ■

Exercise 1.2. Let $(G_i)_{i=1}^n$ be a family of normal subgroups of G such that

$$G = \left\langle \bigcup_{i=1}^n G_i \right\rangle. \quad (1.3)$$

Show that G is isomorphic to the direct product of the G_i if and only if one of the following conditions holds:

- 1. For every i ,

$$G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle = \{e\}. \quad (1.4)$$

- 2. Every element $g \in G$ can be written uniquely as a product

$$g = g_1 g_2 \cdots g_n, \quad g_i \in G_i. \quad (1.5)$$

Solution. ■

Exercise 1.3. Let G be a group, and define

$$\Delta = \{(x, x) \mid x \in G\}. \quad (1.6)$$

- (a) Show that Δ is a subgroup of $G \times G$.
- (b) Prove that Δ is a normal subgroup of $G \times G$ if and only if G is abelian. Moreover, in this case, show that the quotient group $(G \times G)/\Delta$ is isomorphic to G .

Solution. ■

Exercise 1.4. A subgroup M of a group G is called *maximal* if $M \neq G$ and for every subgroup L of G ,

$$M \subsetneq L \subseteq G \implies L = G. \quad (1.7)$$

- (a) Let K, M be subgroups of a group G such that $K \trianglelefteq G$ and $K \subseteq M$. Prove that M is a maximal subgroup of G if and only if M/K is a maximal subgroup of G/K .
- (b) Prove that in a finite group G , every proper subgroup $H \neq G$ is contained in some maximal subgroup of G .
- (c) Prove that in a finite abelian group different from the trivial group, the maximal subgroups are precisely those of prime index.
- (d) Is the previous point true if we remove the assumption that the group is abelian?

Solution. ■

Exercise 1.5. Let p be a prime number.

- (a) Determine all elements of order dividing p in S_n . Denote by a_n the number of such elements and derive a formula for a_n .
- (b) Prove the identity

$$\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n = e^{x + \frac{x^p}{p}}. \quad (1.8)$$

Solution. ■